

Unit 4, Lesson 10: Comparing and Performing Operations with Absolute Value

Benchmark

MA.6.NSO.1.4 Solve mathematical and real-world problems involving absolute value, including the comparison of absolute value.

Learning Targets

- ☐ I can compare the absolute value of two numbers.
- ☐ I can perform operations that involve the absolute values of two numbers.

4.10.1 Warm-Up: Rational Numbers in the Real World

For each of the following problems, ask a classmate for his/her explanation. Once you have received their explanation, write a summary in the space provided in the box. Your classmate should read your summary and initial that it is correct.

Repeat this with a different classmate for each of the remaining problems.

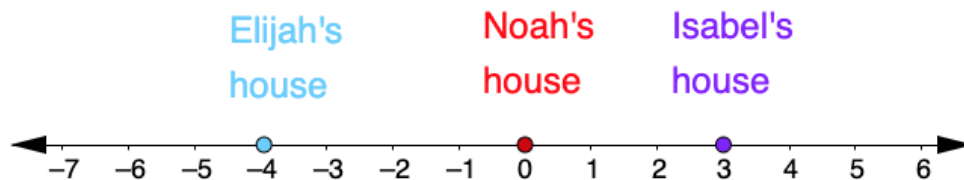
1. Describe the relationship between two numbers that have the same absolute value.

Initials: _____

2. In an American football game, the offensive team makes gains or losses on each play. Gains are represented by positive values and losses by negative values. What does 0 represent in this context?

Initials: _____

3. Elijah, Noah, and Isabella live on the same street. The locations of their houses are modeled on the number line.

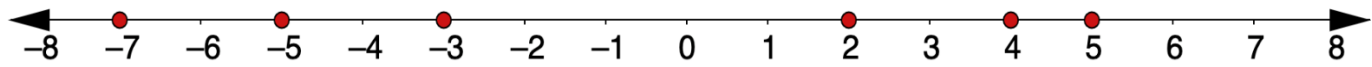


Describe where they live relative to each other in two sentences. Include the words distance and absolute value.

Initials: _____

4.10.2 Guided Instruction: Comparing Absolute Values

A number line is shown with the integers $-7, -5, -3, 2, 4,$ and 5 plotted.



1. Discuss with your partner which statements are true. Be prepared to explain your thinking.

Mathematical Statement	Explanation
$-7 < -3$	
$ -7 $ is 7	
$ -3 $ is 3	
$ -7 < -3 $	

2. Does the comparison of -7 and -3 match the comparison of $|-7|$ and $|-3|$?
3. With your partner, discuss why comparing two negative values and comparing the absolute value of the same two negative numbers may be different. Summarize your discussion below.

4. For each row, use the symbols $>$, $<$, or $=$ to compare each pair of values.

	Inequality	Absolute Value Inequality
Pair 1	$4 \underline{\hspace{1cm}} -5$	$ 4 \underline{\hspace{1cm}} -5 $
Pair 2	$-3 \underline{\hspace{1cm}} 2$	$ -3 \underline{\hspace{1cm}} 2 $
Pair 3	$6 \underline{\hspace{1cm}} -6$	$ 6 \underline{\hspace{1cm}} -6 $

5. What do you notice about the comparisons?
6. When comparing the absolute values of two numbers, numbers that are closer to zero will always have the _____ absolute value.

4.10.3 Collaboration: Inequality Mix and Match

Work with your partner to complete the table.

Partner A will select two numbers and one comparison symbol. At least one of the numbers must be written as an absolute value.

Partner B will then write the mathematical statement in words. The following phrases may be helpful: is equal to, the absolute value of, is greater than, is less than.

Switch roles until each partner has taken two turns. An example is shown in the first row.

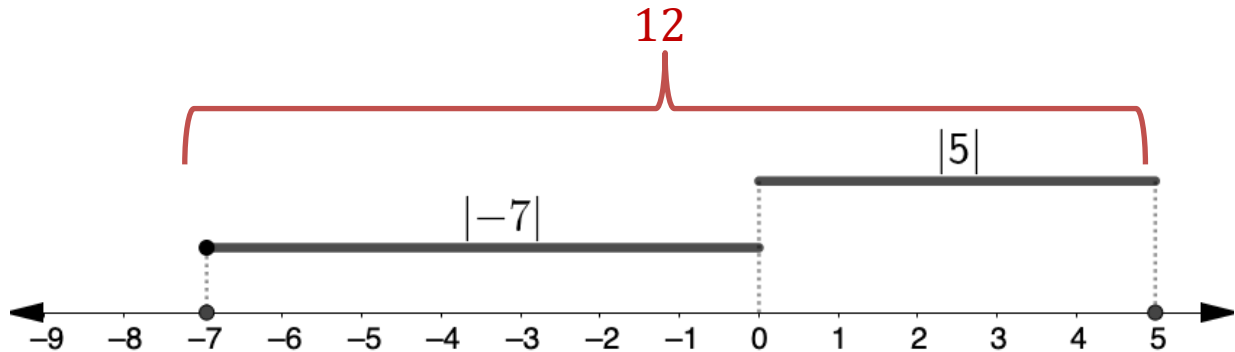
Numbers					Symbol
-0.7	$ -9 $	11	14	$ -8 $	$<$
$-\frac{6}{2}$	-2.2	$ 2.5 $	8	$ 0.7 $	$=$
$ -14 $	$ 20 $	$\frac{7}{2}$	$ 15 $	$ \frac{5}{2} $	$>$

Partner	Mathematical Statement	Statement in Words
Example	$ 3 < 4$	<i>the absolute value of 3 is less than 4</i>
Partner A		
Partner B		
Partner A		
Partner B		

4.10.4 Guided Instruction: Operations Using Absolute Value

1. Ty'shaun and Gabby were asked to explain their strategies for determining the sum of $|-7|$ and $|5|$.

Ty'shaun modeled his thinking on the number line.



Gabby's explanation was "I know $|-7|$ is a distance of 7 from 0 and $|5|$ is a distance of 5 from 0. The sum of 7 and 5 is 12, so the sum of $|-7|$ and $|5|$ is also 12."

Explain which strategy makes more sense to you.

2. Find the value of each expression.

a. $|-18| - |-13|$

b. $12 \times |-4|$

c. $\frac{|45|}{|-3|}$

d. $|-22| + |31| + |-40|$

4.10.5 Your Turn!

1. Use the symbols $>$, $<$, or $=$ to compare each pair of values.

a. $|-42|$ ____ -25

d. $|-512|$ ____ 284

b. -21 ____ $|18|$

e. $|-176|$ ____ $|176|$

c. $|-582|$ ____ $|-790|$

f. $|-38|$ ____ 0

2. Find the value of each expression.

a. $|-58| + 12 =$ _____

b. $|-13| \times 5 =$ _____

c. $\frac{|-36|}{|9|} =$ _____

4.10.6 Wrap-Up: Reflection

Place an **X** on the line to indicate your understanding of each of today's learning targets. Then provide a brief explanation or evidence to support your self-assessment.

1. I can compare the absolute value of two numbers.

I need help.
I can't get started.

I'm getting there.

I understand
and have evidence.



2. I can perform operations that involve the absolute value of two numbers.

I need help.
I can't get started.

I'm getting there.

I understand
and have evidence.

